

# Distributed Intrusion Detection System in a Multi-Layer Network Architecture of Smart Grids

Yichi Zhang, Lingfeng Wang, Weiqing Sun, Robert C. Green II, and Mansoor Alam

**Abstract**—The advent of the smart grid promises to usher in an era that will bring intelligence, efficiency, and optimality to the power grid. Most of these changes will occur as an Internet-like communications network is superimposed on top of the current power grid using wireless mesh network technologies with the 802.15.4, 802.11, and WiMAX standards. Each of these will expose the power grid to cybersecurity threats. In order to address this issue, this work proposes a distributed intrusion detection system for smart grids (SGDIDS) by developing and deploying an intelligent module, the analyzing module (AM), in multiple layers of the smart grid. Multiple AMs will be embedded at each level of the smart grid—the home area networks (HANs), neighborhood area networks (NANs), and wide area networks (WANs)—where they will use the support vector machine (SVM) and artificial immune system (AIS) to detect and classify malicious data and possible cyberattacks. AMs at each level are trained using data that is relevant to their level and will also be able to communicate in order to improve detection. Simulation results demonstrate that this is a promising methodology for supporting the optimal communication routing and improving system security through the identification of malicious network traffic.

**Index Terms**—Artificial immune system, cybersecurity, data classification, intrusion detection, smart grid, support vector machine, wireless mesh networks.

## Nomenclature:

|       |                                   |
|-------|-----------------------------------|
| AIS   | Artificial immune system.         |
| AMI   | Advanced metering infrastructure. |
| AM    | Analyzing module.                 |
| CAC   | Central access controller.        |
| CM    | Controlling module.               |
| DSM   | Data segmentation module.         |
| E&S C | Energy and service corporations.  |
| EDS   | Energy distribution system.       |
| HAN   | Home area networks.               |

|         |                                                         |
|---------|---------------------------------------------------------|
| IAM     | Information acquisition module.                         |
| KDD99   | KDD cup 1999 dataset.                                   |
| MM      | Metering module.                                        |
| NAN     | Neighborhood area networks.                             |
| OM      | Output module.                                          |
| PM      | Preprocessing module.                                   |
| SCADA C | Supervisory Control And Data Acquisition Controller.    |
| SG-DIDS | Distributed intrusion detection system for smart grids. |
| SM      | Service module.                                         |
| SMDC    | Smart meter data collector.                             |
| SVM     | Support vector machine.                                 |
| WAN     | Wide area networks.                                     |
| WMN     | Wireless mesh network.                                  |

## I. INTRODUCTION

THE CONCEPT OF the smart grid promises the world an efficient and intelligent approach of managing energy supply and consumption. Consumers and energy suppliers alike can take advantage of the convenience, reliability, and energy savings provided through real time energy management. One of the advantages of the smart grid is the installation of a completely new, two-way communication network between energy suppliers and their customers. This allows the smart grid to be viewed as a power grid that has an integrated data communication network allowing the collection and analysis of data at all levels in real time. The real-time communication ability of the smart grid will enable utilities to optimize and modernize the power grid in order to realize its full potential [1]. This communication network will provide a number of new energy concepts including real-time pricing, load shedding, consumption management, cost savings from peak load reduction and energy efficiency, integration of plug-in hybrid electric vehicles for grid energy storage, and the integration of alternative distributed generation sources including photovoltaic systems and wind turbines. This new communication network will be constructed using various communication paths including fiber optic cable, twisted pair, broadband over power line, and wireless technologies [2].

The proliferation of these new technologies, especially an Internet-like communications network, may introduce some new threats to the security of the smart grid. Besides cyberthreats like malware, spyware, and computer viruses that currently threaten

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the security of computer communication networks, the introduction of new and distributed technologies such as smart meters, sensors, and other subnetworks can bring new vulnerabilities to the smart grid [3]. In the smart grid network there are three crucial aspects of security that may be threatened due to these vulnerabilities: confidentiality, integrity, and availability (CIA) [4]. For instance, the data flow of the entire network may be interrupted using SYN flood attacks against the metering infrastructure.

Due to these issues, an integrated strategy to improve the security of the power grid with regards to the cyberphysical security of the smart grid is extremely important. In this paper we propose a new architecture for a hierarchical and distributed intrusion detection system called the smart grid distributed intrusion detection system (SGDIDS) that is applied to a representative three-layer communication network. This distributed intrusion detection system (DIDS) is able to successfully analyze communications traffic using an analyzing module (AM) that leverages classification algorithms such as support vector machine (SVM) and artificial immune system (AIS) in order to determine if an attack is occurring, what type of attack it is, and where it comes from in the communication system.

The remainder of this paper is organized as follows: Section II provides a literature review of this research field, Section III presents the architecture and design of the SGDIDS proposed, Section IV gives the simulation results and analysis of the SGDIDS in multiple scenarios, and Section V provides conclusions and future work.

## II. LITERATURE REVIEW

### A. Cybersecurity Issues in Smart Grid

The cybersecurity of the smart grid is quickly becoming an important issue due to the growing awareness of security issues in cyberspace. The potential vulnerabilities and attacks brought about by the inclusion of new technologies that will support the smart grid are discussed in [1], [3], [5], [6].

Several parts of the smart grid may suffer from cybersecurity vulnerabilities including the advanced metering infrastructure (AMI), unsecured wireless mesh network communications, and other attackable technologies.

The AMI is composed of four parts (a smart meter, customer gateway, AMI communication network, and headend) and can generally be considered as a metering system providing two-way communications, automated meter data collection, intelligent outage management, dynamic rate structures, and demand response for load control [4]. It is viewed as a fundamental technology for the smart grid. Currently, there are a number of potential vulnerabilities that have been discovered in AMI networks. For instance, the device memory in the AMI network could be modified by inserting malicious software [7] or the disconnect command may be sent to the meters which would block the transmission of metering information [4].

Wireless networks are also commonly used in the current smart grid because of their convenience and low cost. Some smart grid implementations use mesh networks with wireless

devices to provide self-adapting, multipath, or multihop communication between the nodes. Mesh networks provide redundant communication paths as well as multiple communication paths that can compensate for the failures brought on by the breakdown of communication nodes. For instance, Zigbee is a low-cost and low-power wireless mesh networking protocol which can be widely deployed and utilized in a more reliable environment. This makes it advantageous to use the technology in the smart grid even though it allows for easy attacks due to the commoditized and open nature of the wireless radios used. An example of this can be found in [8] where DoS attacks are associated with Zigbee implementations in the smart grid. Further examples of attacks that are used against wireless network technologies include damaging the integrity of configuration, routing, and communication traffic, illegitimate network operations, inconsistent traffic direction, alteration of data, unintentional dissemination of consumer data, unresponsive destination nodes, grid bandwidth over-usage and signal power over-consumption.

Each two-way communication path which supports control and measurement in the smart grid also has the potential to become an entry point for both physical and cyberattacks that may be used by anyone with mal intent. Wireless networks can easily be probed by attackers and are susceptible to man-in-the-middle (MIM) attacks. Although there are security mechanisms which could prevent unauthorized use of these communication paths, weaknesses still exist in these mechanisms. What is more, in smart meters and supervisory control and data acquisition (SCADA) systems, it is possible to log on to these nodes and re-program the measuring and controlling commands. This could lead to significant errors in power measurements which could, in turn, lead to severe power outages. Also, without a fixed standard or proprietary security mechanism, the implementation of the smart grid may lead to poor interoperability and issues of security by obscurity. This means that if the security of the smart grid is designed based on the concept of hiding all of the known vulnerabilities and flaws, knowledgeable attackers may overcome network security and the grid with ease [9]. Although a series of regulations for critical infrastructure protection (CIP) have been published, a mature standard is still in process.

In the portion of the power system dealing with monitoring and control, a number of digital vulnerabilities have been discovered at the entrances to substations of the SCADA systems [10]. For instance, multiple access points of the SCADA could be attacked. Examples of this include blocking or probing the communications between the modem and the RTU. A DoS attack is simulated in [11] where clients communicate with the simulation of a power system that was implemented using PowerWorld. The work in [12] also discusses the possible cyber vulnerabilities and their detection regarding the SCADA system.

Currently, many distributed intrusion detection systems have been discussed in the literature, much of which is focused on applications in the area of the computer networks. For instance, in [13] a distributed community based intrusion detection system was proposed. This system was applied to detect security threats known as worms. The work in [14] proposes a dynamic, hierarchical, multiagent-based DIDS that is designed to detect distributed attacks in communication networks. The work in [15]

proposes a network-based DIDS that leverages an intrusion detection message exchange format to support digital signatures and increase the confidentiality of the entire communication system. An intrusion detection system which uses autonomous modules was proposed to monitor the security of the networks in [16]. These models are successfully applied but limited to pure computer networks.

### B. Intrusion Detection in Smart Grid

Since the smart grid network is a hybrid of the power system and a communication network, intrusions should be detected that concern either the physical power system or the communication network or both. The combination of these issues refers to the cyberphysical aspect of the smart grid. In this paper, cybersecurity of the cyberphysical power system is addressed, which considers the cybercomponents from both cyber and physical domains. For instance, phasor measurement units and FACTS devices can be deemed as physical (hardware) components, but their cyber parts (e.g., embedded software) also can be intruded, which will lead to various network traffic patterns. Our cybersecurity study also covers the cyber vulnerabilities from these physical devices which include both hardware and software components.

In the SGDIDS, it is crucial to choose an efficient communication topology and a reasonable communication standard as well as a robust attack classification algorithm. In light of this, many works about network topologies, standards, and classification algorithms were reviewed. In [2] and [17] the wireless mesh network topology was discussed and compared with other topologies. It was considered that the mesh network could supply a reliable, robust, and cost-effective topology for communication. In [18] the construction and parameters of wireless mesh networks in different environments were discussed. In [19] and [20] threats regarding the security of the wireless networks were presented and some solutions, such as secure routing protocols, were proposed.

The standards which are suitable for the wireless mesh topology are also studied. Zigbee is a low-cost and low-power network standard which can be widely deployed and utilized in the limited range of wireless mesh network environment [21]. The 802.11n Wi-Fi standard can enable a further range of communication with faster data transmission speeds than was previously cost prohibitive or impractical in wireless mesh networks [22]. WiMAX can also provide low costs and remote communication within the wireless mesh network [23].

One of the requirements of the SGDIDS method proposed is the ability to classify attacks efficiently and effectively through the use of a robust classification algorithm. In order to accomplish this, a support vector machine (SVM) is chosen as the classification algorithm because of its convenience of usage and high accuracy in classification [24]. Many experiments have also applied SVM algorithm on the KDD99 dataset [25] for data mining. In [26] an enhanced SVM model was created based on the important features of the training data. In [27] a radial basis function (RBF) was used as the kernel for the SVM algorithm in order to classify the KDD99 dataset. The work in [28] and [29] presents a methodology for applying SVM to the multiple classification of a large, scaled dataset. A popular SVM tool named

LIBSVM [30] is applied for the classification of the intrusion data in this work as it provides good computational performance regarding attack detection.

In order to make a comparison with SVM algorithm, we also apply the artificial immune systems (AIS) as the classification algorithm. AIS are computational algorithms that emulate the mechanisms of human immune systems [31]. They involve learning, memory, and optimizing capabilities for conforming supervised and unsupervised computational algorithms. The primary advantages of AIS are that only positive examples are needed in the algorithms, and the patterns AIS has been trained with or learned can be explicitly examined. There are mainly four fields of algorithms which are derived from AIS: negative selection algorithms, immune network algorithms, danger theory, and clonal selection algorithms. In this application, the clonal selection algorithm is considered because of its flexibility. Its theory is used for emulating the basic process of an adaptive immune response to the antigenic stimulus. Only those cells that can recognize the antigens are allowed to clone and proliferate.

The clonal selection theory was first proposed by Burnet in [32] and numerical research has been completed by other scholars. For instance, de Castro proposed the clonal selection algorithm (CSA) algorithm in [33], transformed it into the CLONALG algorithm in [34], the clonal selection classification algorithm (CSCA) was reported in [35], and in [36] a dynamic clonal selection algorithm with the property of self-adaptation was proposed. In [37] Watkins proposed the artificial immune recognition system (AIRS) algorithm which competes with CLONALG in terms of efficiency. The improvement of parallelization was described in [38].

## III. SYSTEM DESIGN

The method that we propose for integrating greater cybersecurity into the smart grid communication scheme consists of three essential pieces: A three-layer network infrastructure with multiple distributed modules, the use of SVM/AIS for attack classification, and an optimal communication scenario among distributed modules.

### A. Network Architecture

When considering the large communication network that will exist in the smart grid, we propose a three-layer network composed of home area networks (HAN), neighborhood area networks (NAN), and wide area networks (WAN) that may be assumed as a reasonable infrastructure design.

The first layer of the network is the home area network (HAN) which consists of the service module (SM), the metering module (MM), and the HAN intrusion detection system (IDS) module. The SM will provide real-time energy cost and consumption data to the consumers while the MM (which includes the smart meters) will record the consumption of the energy in a consumer's home. The HAN IDS will, as expected, track and scrutinize both incoming and outgoing communications in order to determine if any lapses in security are occurring.

The second layer is the neighborhood area network (NAN). The NAN is a large metering and controlling network which collects metering and service information from the multiple HANs

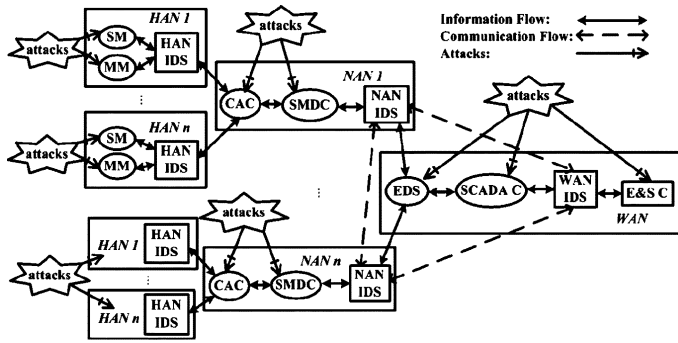


Fig. 1. Three-layer network architecture.

that are geographically near each other. The NAN will consist of three components: the central access controller (CAC), the smart meter data collector (SMDC), and the NAN IDS. The CAC can be considered as the interface that manages the communication between HANs and the energy supplier or utility. The SMDC will be a wireless node that is in charge of the metering record of the whole community as composed by the neighboring HANs. All data that is flowing either in or out of the NAN will be passed through the NAN IDS in order to detect any possible security threats.

The final layer is the WAN. It provides broadband wired and wireless communication between the NAN, substations, other distributed grid devices, and the utility. This layer will consist of the energy distribution system (EDS), the SCADA controller, and the WAN IDS. The EDS will remain in charge of the energy and metering data distribution. The SCADA controller provides the utility or grid operator with distributed process control in order to manage the distribution grid elements. Since the large-scale control and metering data are crucial to the energy and service corporations (E&SC), a WAN IDS is required between the SCADA controller and the supplier in order to maintain security. A graphical depiction of all three levels of the proposed network can be seen in Fig. 1.

Because of the recent implementation of standards-based wireless communication protocols such as IEEE 802.15.4 Zigbee, the widespread use of 802.11 protocols, the use of 802.16 WiMAX, and the need for distributed sensing technology, it can be assumed that large parts of each network layer will exist in the form of a wireless mesh network [2].

The wireless communication network topology chosen for the SGDIDS architecture is the wireless mesh network (WMN). The mesh network topology is adopted and is beneficial due to its redundant communication paths which compensate for natural failure as well as its scalable, dynamic, and self-healing capacities [2]. Mesh networks also provide multiple, redundant communication paths that can compensate for the failures brought on by the breakdown of communication between nodes due to failures, congestion, and other maladies. It is known that the majority of utilities have applied the wireless mesh system for smart grid components. The earth2tech's Katie Fehrenbacher estimated that 96% of the market is owned by wireless mesh networks when smart grid network installations are considered [39].

Since thousands of network standards are able to support the smart grid protocols and networks, requirements should be set

for selecting the optimal network standards for the communication network of the smart grid, though these requirements will differ across the different layers. In the lower HAN and NAN layers the requirements include high network capacity and data transmission rate. In the third layer, stability and reliability of the communication network is the main requirement.

In the HAN, the network standard chosen is 802.15.4 Zigbee. This protocol provides radio communication specifically suitable for personal and home area networks [8]. It has three types of bands: 868–868.6 MHz with 1 channel for European utilities, 902–928 MHz with 10 channels for North American utilities, and 2.4 GHz with 16 nonoverlapping channels for worldwide utilities. The 2.4 GHz band is chosen since it supports the data rate of 250 kbps as over-the-air data transmission which is higher than the data rates of other two bands. This band also has 16 channels. The maximum wireless capacity is calculated in (1) where  $C_{\max}$  represents the maximum wireless capacity,  $r$  represents the data rate, and  $N_C$  means the nonoverlapping channels. Thus, the maximum capacity of one data link under the 802.15.4 Zigbee standard is 4 Mbps

$$C_{\max} = r \times N_C. \quad (1)$$

In the NAN, 802.11n is chosen as the communication standard. Compared with other Wi-Fi standards, 802.11n increases the over-the-air data rates to a maximum of 300 Mbps on the 2.4 GHz band. This band is also shared with 802.15.4 Zigbee in the PHY layer [40]. The number of possible un-overlapping channels achieved is 23. The communication range is the largest when compared with the other 802.11 standards as the total range can be up to 250 m outdoors. By using (1), the maximum wireless capacity of links under the 802.11n standard is 6900 Mbps.

In the WAN, since the elements are mostly fixed and physically distant from each other, both wireless and wired networks will be applied to assist the availability and reliability between the communications of the smart grid elements. In our network, the large amount of data that is transmitted is sent via wired communication techniques such as power line communication (PLC), while the control information is transmitted by a wireless network. The wireless standard chosen is the WiMAX 802.16-2009. This protocol is currently utilized with a 70 Mbps data rate under the spectrally efficient orthogonal frequency division multiplexing (OFDM) mechanism. The most crucial reason for choosing WiMAX is that it can provide communication at a distance as large as 2 miles [23]. By applying the WMN topology and maintaining high data rate standards, the SGDIDS can provide a robust communication environment that is also congestion aware. For instance, when one access point of a smart meter is broken or flooded with traffic, the user can still read their metering information by accessing data from the other communication nodes or components. Also, by applying the broadband and high capacity wireless standards, SGDIDS is able to satisfy the requirement of transmitting a large amount of data.

Though the use of wireless communication technologies is a necessary aspect of the smart grid, these technologies introduce new vulnerabilities and security issues into the smart grid. Because of this, it is crucial to construct a cybersecurity strategy to

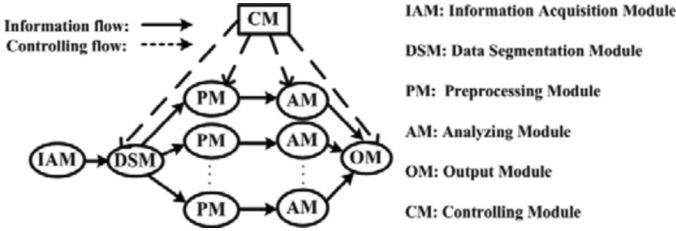


Fig. 2. Home area network intrusion detection system.

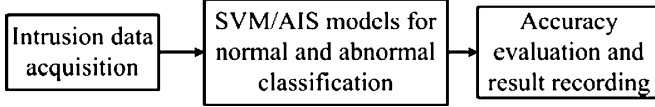


Fig. 3. Analyzing module architecture.

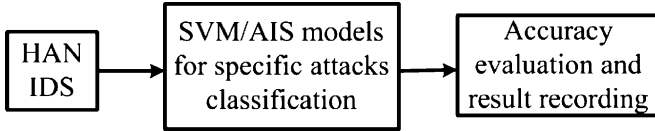


Fig. 4. Neighborhood area network intrusion detection system.

protect the confidentiality, integrity, and availability of the data transmission in the smart grid. The inclusion of IDS technology provides a process of identifying the network activity that may lead to a compromise of security policy. It also monitors activities from access points while recording and preventing the suspicious activities which are marked as intrusions [26].

### B. Distributed IDS Modules

The HAN IDS is designed by combining multiple intelligent modules as shown in Fig. 2. The information acquisition module (IAM) collects the data packets of the energy consumption information and saves them in a matrix. The data segmentation module (DSM) partitions the received data into proper-size segmentation files which are suitable for the detection algorithm with the preprocessing modules (PM). The analyzing modules (AM) are used to detect suspicious intrusions. An AM is composed of three parts as shown in Fig. 3. The intrusion data acquisition component saves the processed data from the PM and the intrusions are classified by using the trained SVM algorithm or the AIS algorithm. Finally the intrusion types, attack time, and the addresses are recorded and printed out by the output module. The whole detection process is, in the HAN, controlled by the controlling module (CM).

The IDSs that are designed for NAN IDS and WAN IDS are shown in Figs. 4 and 5. Each one is a combination of the best HAN IDS and other classifier models focusing on a better classification of the specific malicious attacks through the processing of massive amounts of data. For the attacks which are difficult to be classified in the current layer, the evaluation results will be checked in order to determine whether the data should be sent to the higher layer for further evaluation. Also, in WAN IDS, a central controller is needed for controlling the NAN IDS in the sublayer.

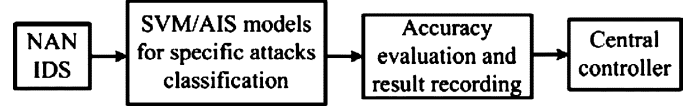


Fig. 5. Wide area network intrusion detection system.

### C. Wireless Mesh Network Model and the Optimal Routing Algorithm

Here we use a directed graph,  $G = (V, E)$ , to illustrate the multihop multichannel network [42]. Each node  $V$  can be considered as a smart grid component for measuring electricity consumption and managing communications with other nodes or higher layer components. Each edge,  $E$ , represents the wireless links between the nodes. In this graph interference amidst communications is not considered for simplicity. Only  $C(e) \in C$  channels can be activated on any link  $e$  of the network where each channel is orthogonal to another. The data sent on each channel is transmitted by one and only one radio,  $i$ . Each channel of the data link has a capacity of  $c_i(e)$ . This is a constant showing the maximum data flow on that channel. The information flow currently transmitting on channel  $i$  of link  $e$  is denoted as  $f_i(e)$ .

In order to make sure the data flow can be transmitted in the channels smoothly, the constraints for the link or channel and all flows should be set appropriately. Suppose the scheduling variable  $x_i^t(e)$  exists as  $x_i^t(e) = 1$  when data is transmitting on link channel  $i$  of link  $e$  during time slot  $t$ . Thus, the maximum number of the active channels is  $C(e)$  during time slot  $t$  is

$$\sum_{i \in C} x_i^t(e) \leq C(e) \quad e \in E. \quad (2)$$

Also, we have  $\lambda$  sets of link and channel constraining pairs  $(S_1, S_2, \dots, S_\lambda)$  between each pair of nodes which are allowed to communicate to each other directly. In each pair,  $S_j$  means the constraining set belongs to channel  $i$  of link  $e$ . Each pair has a RHS constant,  $\beta(S_j)$ , according to the following constraint:

$$\frac{1}{\beta(S_j)} \sum_{(e,i) \in S_j} \frac{f_i(e)}{c_i(e)} \leq 1, \quad j \in \{1, 2, \dots, \lambda\}. \quad (3)$$

The nodes communication's with each other consist of  $M$  source destination pairs (shorted as commodity pairs) denoted as  $(s(m), d(m))$ . In each pair the flow allowed on channel  $i$  is set to be  $p_i^m(e)$  and the total flow,  $r(m)$ , transmitted between the pair can be calculated as  $\sum_{e=s(m)} \sum_{i \in C} p_i^m(e) = r(m)$  according to the constraint

$$\frac{1}{\beta(S_j)} \sum_{S_j} \frac{\sum_{m \in M} p_i^m(e)}{c_i(e)} \leq 1. \quad (4)$$

The primal-dual algorithm is applied to find the optimal path for communication flows. Set  $SF$  as the maximum scaling factor where its value denotes the total slack capacity needed in the network. Because of the constraints above, it is required that  $SF$  should not be smaller than 1. The primal-dual algorithm is described as Fig. 6.

In this algorithm  $w(j)$  is the weight of the constraining pair of link  $i$  channel  $e$  and  $\alpha(j)$  is the weight of each set  $S_j$  whose

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 $w(j) = \delta, j \in \{1, 2, \dots, \lambda\}$ 
 $b = 0$ 
While  $\sum_{j=1}^{\lambda} w(j) < 1$ 
  For  $m = 1, 2, \dots, M$ 
     $r = r(m)$ 
    While  $r > 0$ 

      Set  $\Delta w(j) = \frac{\sum_{j \in S, \beta(S_j)} \alpha(j)}{c_i(e)}$ 
       $P_{\min}(s(m), d(m)) = \min_{m \in M} \sum \Delta w(j)$ 
       $\mu = \min(f(P_{\min}))$ 
       $\sigma = \min\{r, \mu\}$ 
       $r = r - \sigma$ 
       $f_i(e) = f_i(e) + \sigma$ 
       $w(j) = w(j) * (1 + \frac{\varepsilon \sigma}{f(P_{\min})})$ 
    endwhile
  endfor
   $b = b + 1$ 
endwhile
 $g = \max \left( \sum_{j \in S} \frac{f_i(e)}{c_i(e)} \right)$ 
 $SF = \frac{b}{g}$ 

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Fig. 6. Primal-dual algorithm [42].

value is calculated by the distance between two nodes. When searching for the optimal route, all the routes starting from the source node and ending with the destination node will be counted and the accumulation of the constraining pairs' weights will be calculated. Then the route which has the lowest value of weight accumulations will be labeled as the shortest path. The desired data flow,  $r(m)$ , is distributed to the corresponding node, and this source destination pair, which is also called a commodity pair, can find the optimal route with the shortest distances and link weights. By using this algorithm, the problem of finding the optimal path and the flow distribution is settled. Through the application of the primal-dual algorithm to find the shortest path of communication, a communication network could be provided to satisfy the requirement of efficient power consumption in the smart grid.

#### D. SVM for Attack Classification

Machine learning via SVM has previously been found to exhibit improved performance and is a powerful tool for the classification of data [43]. SVM is a type of machine learning technique that attempts to successfully classify sets of data by leveraging two basic principles: large-margin separation and kernel functions. Large-margin separation refers to the idea that when classifying data it is sensible to draw a line of separation in such a manner that the distance between that line and the closest data point on either side of that line is maximized. Kernel functions are algorithms or functions that calculate the similarity between two points and must be used if nonlinear classification boundaries are required. Generally speaking, as the dimension of classification increases hyper planes are used to separate the data

instead of lines. This is completed by mapping the data to a different space (also using a kernel function) where a hyper plane is able to classify the data.

In order to achieve a classification based on these ideas, a convex, quadratic program must be solved that is, in the case of data that is nonseparable, of the form

$$\min \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1}^l \xi_i \quad (5)$$

$$\text{such that } y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i \quad (6)$$

$$\xi_i \geq 0 \quad (7)$$

where  $w$  is the weight vector,  $C$  is a value that controls the variance between margin maximization and error minimization,  $\xi_i$  is a set of slack variables that measure constraint violation,  $y_i$  denotes that there is a unique constraint for each necessary classification  $i$ ,  $b$  is the bias,  $x_i$  is a training vector, and  $\phi(x_i)$  is the kernel function that maps the training or testing vector to a different space [44], [45]. The formulation is the equivalent of maximizing the margin while also minimizing error.

Often it is also necessary to classify data into three or more categories. Two methods are typically used to perform this classification: the one-against-many and the one-against-one approach. The former approach is simpler and examines one category at a time while merging the remaining categories into a single category. This allows for simple, binary classification. The latter approach is more computationally intensive though more accurate as it creates  $k(k-1)/2$  models in order to classify all data where  $k$  is the number of classifications required.

#### E. Clonal Selection Classification Algorithm for Attack Detection

The clonal selection principle describes the process of the reaction between the immune system and antigens. During the reaction, lymphocyte cells are created so that the unknown antigen is better recognized. The cells start their proliferation through cell cloning and differentiation that is based on mutation. The activation of the B-cell evaluates its binding to the antigen. The number of clones produced by the activated B-cell depends directly on its level of activation. The higher level the activation has, the more clones will be produced. This fact leads to faster immune adaptation since B-cells with low affinity are differentiated through high mutation rates. This may also increase the affinities of the following generations. The B-cells with poor affinities after the differentiation process are effectively eliminated. The elimination of less qualified B-cells and the affinity-proportionate cloning of B-cells emulate the natural selection process, meaning that only the cells with highest adaptation to the environment are selected and regenerated [46].

In the area of AIS there has been evidence that the nature of the immune system can be realized and applied in network intrusion detections [47]. In this paper we apply two clonal selection algorithms, CLONALG and AIRS2Parallel, for the generation of classifier models.

1) *CLONALG Algorithm*: The CLONALG algorithm is successfully applied in many complex problems with a low complexity structure. When being applied in classification problems it provides promising accuracy. CLONALG was proposed by

De Castro, named as the clonal selection algorithm (CSA), and later renamed as CLONALG. This algorithm takes a population of antibodies and then creates a population which is more sensitive to the antigens by exposing the antibodies to the antigens for several generations. As described in [33] and [35], the basic CLONALG algorithm achieves this by repeating the maintenance, selection, cloning, and mutation of the antibodies until they reach a threshold of sensitivity to the antigens.

Compared with the AIRS algorithm, CLONALG has a lower complexity and a smaller number of parameters that influence the classification accuracy.

The main algorithm for a single generation of CLONALG is described as follows:

1. Randomly generate an initial population of antibodies ( $A_p$ ), which is composed of two subsets: A pool of the memory cells ( $M_p$ ) and the reservoir pool ( $R_p$ ). Also create a set of antigenic patterns,  $A$ ; and
2. Select one antigen  $A_i$  from  $A$ , measure the affinity between  $A_i$  and every member of  $A_p$ , and pick  $n$  antibodies which have the highest affinities;  
Affinity is the value of the similarity between two sets of data. A simple approach to measuring affinity is the Hamming distance between two sets of strings with equal length. The Hamming distance is calculated by using (8)

$$D = \sum_{i=1}^m \eta \quad \text{where } \eta = \begin{cases} 1 & A_{pi} = A_i \\ 0 & A_{pi} \neq A_i \end{cases} \quad (8)$$

where  $D$  is the Hamming distance,  $A_{pi}$  is the antibody,  $A_i$  is the antigen, and  $m$  is the length of the data string.

3. Regenerate the clones of the  $n$  antibodies with the highest affinities and place them in a new population  $C$ . The number of clones is decided by the values of the affinities;
4. Mutate the clone population  $C$  at a rate inversely proportional to their affinity to produce a mature population  $C'$ ;
5. Re-evaluate the affinity to each member of the population  $C'$  and select the antigen with the highest affinity as the candidate memory cell of the memory cell set  $M_p$ . The member of the memory cells is updated when the affinity of  $C_i$  is greater than the current  $M_{pi}$ ; and
6. Replace the antibodies with low affinities in  $R_p$  with those of high affinity. Repeat two to six until all the antigens are involved and present it to the antibodies. This completes one generation of CLONALG is completed.

After  $G$  generations of immune training, a matured memory pool is created as the result of the CLONALG algorithm. The elements in the memory pool can be applied to different problems such as pattern recognition, optimization, and the traveling salesman problem (TSP).

2) *AIRS2Parallel Algorithm*: The artificial immune recognition system (AIRS) algorithm is one of the immune-inspired supervised learning algorithms which include clonal selection, affinity maturation, and affinity recognition balls (ARBs) [38]. In the experiments of [48] it has been concluded that AIRS has the highest value of classification accuracy. The AIRS algorithm is a cluster-based approach especially for data classification that optimizes the location of cluster centers. This is different from CLONALG which is an unsupervised learning algorithm. But

similar to CLONALG, AIRS also focuses on the development of the memory cells over multiple generations. There are five steps in the AIRS algorithm: Initialization, antigen training and competition, memory cell selection, and classification of the dataset. The second to fourth steps are repeated iteratively.

First, the dataset is normalized and the affinity threshold variable is calculated. Then two different populations, the artificial recognition ball (ARBs) pool and the memory cell pool are created. In the antigen training step, each antigen is required to be exposed to the memory pool. The best memory cell for the memory pool is selected when the stimulation value is maximized during the stimulation process. This cell is then applied for the affinity maturation process where it is cloned and mutated based on its value of the affinity. The clones are then added to the ARB pool. The competition then starts and ARBs which have low resources are discarded from the pool until the stopping criterion is satisfied. The ARB which has the highest stimulation value is selected as the candidate memory cell and is added into the memory cell pool. These steps are repeated until all the antigens are tested and then the classification will take place by performing the  $k$  nearest neighbor search in the memory pool [49].

The AIRS2Parallel algorithm may also increase computational efficiency [50]. The difference between this algorithm and the AIRS algorithm is that the dataset used in AIRS2Parallel are departed for the parallel calculation and the combination of the memory cells into a single pool for classification. By using this approach, both computation speed and the accuracy of the classification are increased.

#### F. Dataset and Preprocessing

In the proposed models, the SVM and AIS algorithms will be used as the two methods of attack classification. Each IDS at the HAN, NAN, and WAN level will incorporate these methodologies in order to detect attacks and stave off intrusions. In order to accomplish this, we have incorporated SVM using LIBSVM and CLONALG/Airs2Parallel using the Weka plug-in [51]. These tools are use with an improved and simplified version of the KDD Cup 1999 Data set (KDD99) [25] called the NSL-KDD dataset [52].

The NSL-KDD dataset is a short form of KDD99 which was gathered at MIT Lincoln Labs. KDD99 is the most widely utilized dataset for the evaluation of the anomaly detection as it is an extremely large data set that has a huge number of redundant records of network traffic. However, because of the redundant and repeating records of DoS attacks and the biased distribution of the four types of the attacks, the accurate classification of the U2R and R2L becomes difficult. To address this problem, we use NSL-KDD as the dataset for training and testing. The NSL-KDD dataset discarded the redundant and duplicated records so that the training time is significantly reduced. In total, the training data consists of 125 937 samples consisting of 41 features and 22 separate types of attacks and the testing data composes of 22 544 samples of records. The attacks can be divided into four separate categories including denial of service (DoS), user to root (U2R), remote to local (R2L), and probing/scanning. This dataset can be considered as an abbreviated form

of possible attack scenarios that are applicable when considering the smart grid's communication network.

Smart grid can be considered as a combination of the computer network and the power system with some new characteristics. Thus, some communication scenarios in the current computer network can be applicable to the smart grid. For instance, some attacks occurring in the computer network can find their counterparts in the smart grid such as the block of the communication traffic and the theft of the data. Although the KDD 99 dataset used in the SGDIDS is based on the communication scenarios in computer networks, the features of the record samples such as the service duration time can also be applied to the smart grid communication network scenario. Besides the network traffic patterns, other features of the power system itself (i.e., the current-carrying part) such as the voltage or current values, power flow values, and phasor measurements can be incorporated to enhance the dataset. Based on the variations of power system characteristics under different cyberattacks, the dataset used for classifier training and testing can be developed.

Concerning SVM, our model uses a Gaussian radial basis function for the kernel [44]. The models are trained using a partition of the NSL-KDD dataset. Because of the hierarchical three-layer structure of the SGDIDS, the distribution of training data differs layer by layer. As IDSs in the HAN layer have a limited chance of suffering from most types of attacks, we use training data that has a higher concentration of normal and DoS, probing attacks. The NAN IDSs data differs as its focus is on less common attacks like the R2L attacks. The WAN IDS will focus more heavily on U2R attacks. Because of these differences, training data that have different concentrations of attack types are used at each level.

The training and testing data used is prepared for classification by using the prescaling approach to create a format suitable for use with the SVM methodology and the LIBSVM library [4], [53], [54]. In large-scale data mining it is very important to make the dataset sparse. In order to accomplish this, all nonnumeric data (protocol types, service types, and flag types) are mapped to numeric numerals. As an example, in this study the feature called "attack type" is changed from the values DoS, R2L, U2R, and Probing to the values 1, 2, 3, and 4 respectively. After this step, all numeric data types are scaled to a range between 0 and 1. This is done by dividing each specific value ( $V$ ) by the difference of the maximum ( $F_{\max}$ ) and minimum ( $F_{\min}$ ) value for that feature according to (9). All of these values are then stored in a file in the format specific to LIBSVM

$$\frac{V}{|F_{\max} - F_{\min}|}. \quad (9)$$

In CLONALG and AIRS2Parallel, training and testing data sets used are preprocessed for classification by using the NormalizeMidpointZero filter in Weka for prescaling. This increases the accuracy of the classification. After this step, all numeric data types are scaled to a range between  $-1$  and  $1$ . All of these values are then stored in a file and trained by using the Weka GUI platform.

All simulations are run using Matlab on a computer with Ubuntu 10.04, two Intel Xeon 5504 quad core processors, and 16 GB of RAM.

## IV. SIMULATIONS RESULTS AND ANALYSIS

### A. Distribution of Possible Attacks in Different Layers of SGDIDS

In the smart grid network, the threats launched are different from those found in a typical computer network. In [54], forms and approaches of the attacks in the smart grid scenario are surveyed and discussed. It is found that the specialized layout, design, and communications of the smart grid network expose it to more possible vulnerabilities than a typical computer network. As an example, the AMI may suffer from cyber vulnerabilities related to routing, configuration, name resolution, or authentication protocols while also suffering from physical (hardware) vulnerabilities and encrypted key access.

When compared to the attacks in computer networks, attacks in the smart grid environment should also consider the motivations of its attackers. These motivations can be categorized into five areas: Curiosity for information, motivated attacks, unethical power theft, power consuming information theft, and financial motivations for economic benefits [54].

While these motivations are important, our model considers all attacks and motivations to be encapsulated in the NSL-KDD dataset. These attacks are also illustrated in Fig. 1. Based on literature regarding the cybersecurity of the smart grid, it can be assumed that diverse attacks occur in different layers of the SGDIDS.

In HANs, wireless probing of consumers metering data and the stopping of data transmission between meters may occur. These types of attacks belong to the categories of probing and DoS attacks respectively. In NAN, probing attacks may occur in order to steal the control information of the CAC or to disable the connection between the metering modules and the CAC can be constructed by the DoS attacks; what is more, the data security parameters or controlling information could be reset or altered by the U2R attacks. In WAN, besides the DoS attack as disruption of the metering data transmission to the EDS or E&S C, and the probing attack as large-scale disclosure of customer's metering data, the components in WAN may also suffer from the U2R and R2L attacks such as the alteration of metering and controlling data in the EDS and E&S C, or broadcasting of control commands from the SCADA [1], [54]. In a word, the types of attacks increase in the higher layers of the architecture.

The IDS that is placed into the communication network will detect the existence of and classify the types of these intrusions.

### B. Grid Topology and Communication Routing Optimization

The wireless mesh communication scenario that uses SGDIDS is simulated using MATLAB as it is a powerful tool for simulating a complex communication system. Fig. 7 shows the simulation of the hierarchical, three-layer smart grid communication network proposed in Fig. 1. As can be seen, the network is composed of components by nodes with nodes 1 to 20 in the HAN, nodes 21 to 23 in NAN, and node 24 in the WAN. In order to ensure the security of each component, every node has an IDS belonging to its corresponding network. It means the HAN IDSs are embedded into nodes 1 to 20 in the HAN, and the NAN IDS and WAN IDS are combined in the nodes 21 to 23 and 24 respectively in NAN and WAN.

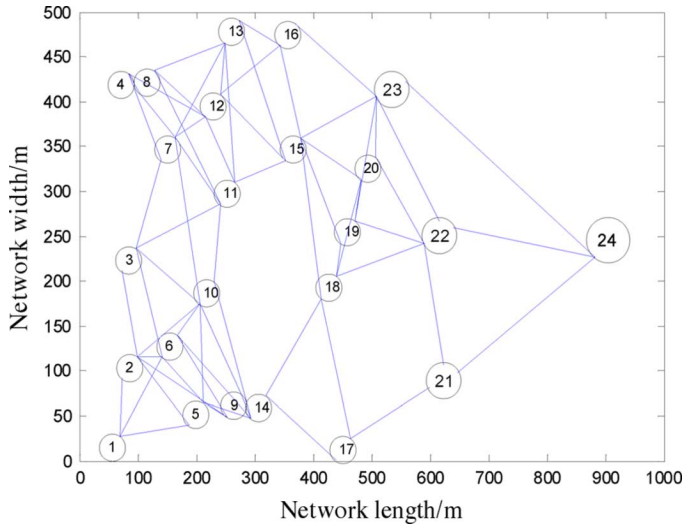


Fig. 7. Smart grid distributed intrusion detection system topology.

In Fig. 7, a mesh network with 20 HAN nodes which are scattered randomly in a  $500 \times 500$  square is generated. These nodes representing the SM or MM compose several HANs. According to the rules of the communications among the nodes in a wireless mesh network, a node can directly communicate with another node by the link between them as long as it is located less than a specified distance away. Here the direct communication distance as set to about 100 m since the longest communication distance of Zigbee is 150 m. Remote node pairs that need to communicate with each other will use the multiple-hop approach by sending their communications through other nodes. The medium sized nodes (21 to 23) represent the CAC or SMDC components in each NAN. These components can directly communicate with the nearest nodes in HAN by using the 802.11n standard. Remote nodes in HAN that need to communicate with other nodes in the NAN can send the information through other HAN nodes. Since one NAN node may route the metering and detection information from multiple HAN nodes, the capacity of the each node and link in the NAN is set higher.

Adding the largest node (24) into the WAN completes the entire  $1000 \times 500$  network. The WAN node can be considered as an EDS, a SCADA Controller, or the E&SC. As the communication between the WAN node and NAN node is completed across a larger distance, the majority of power for communication is consumed at the intermediate nodes. As such, it is necessary to make sure that there is enough power available at these intermediate nodes. The wireless communication standard between a NAN node and the WAN node uses the WiMAX protocol. The link between a WAN and NAN has only one wireless channel in this study. The wired communication standards are applied for metering data transmission because of the larger distance and the great amount of data.

The source and destination node pairs are randomly picked from HAN, NAN, and WAN nodes. The source nodes are tasked to send out packets to the destination nodes directly or by using multiple hops. In this simulation, five commodity pairs are set with each pair labeled using one color. The corresponding route

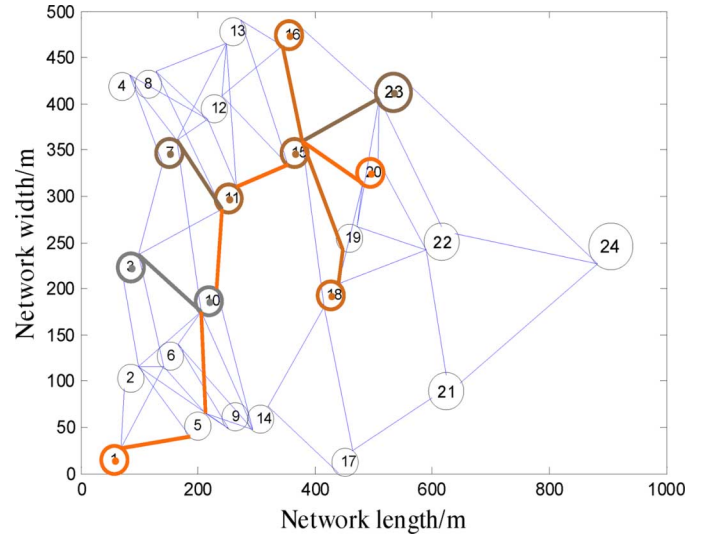


Fig. 8. Optimal communication routing selection in the SGDIDS.

TABLE I  
THE ROUTING TABLE OF THE COMMUNICATION AMONG NODES

| Source node | Intermediate nodes |    |    |    | Destination node |
|-------------|--------------------|----|----|----|------------------|
| 3           |                    |    |    |    | 10               |
| 7           |                    |    | 11 | 15 | 23               |
| 11          |                    |    |    |    | 15               |
| 16          |                    |    | 15 | 19 | 18               |
| 20          | 15                 | 11 | 10 | 5  | 1                |

is also labeled with the same color that is used by the commodity pair. If the node is both used as the source and the destination, its color would be overlapped. The shortest routing for the commodity pair is found by using the primal-dual algorithm described in Section III-C. In order to more realistically simulate the data transmission, the number of channels and the data rate is randomly set on each link under the rule of the network constraints. This means that in the HAN a maximum 16 channels can be opened simultaneously on one link. In the NAN the highest number of channels is 23. Fig. 8 illustrates the communication routing between the commodity pairs and the optimal routings are recorded in Table I.

In Table I, it can be seen that the route chosen by commodity pair (16, 18) uses nodes 15 and 19 as hopping nodes. Although the distance between node 15 and 18 is shorter than the distance from 15 to 19 and 19 to 18, the link between 15 and 18 has a higher weight value since fewer channels are open on this link. This is an example of how using only the shortest path may bring about communication congestion. Thus, by applying the primal-dual algorithm, optimal routing of the communication could be achieved.

### C. Intrusion Detection With the SGDIDS

The detection result is presented in Fig. 9. When the packets arrive at the destination node, the destination node will use its IDS to extract the 41 features of the KDD99 from the communication. Some features that are examined include the duration of the node's connection and the protocol type of the connection. The DIDS will then use its classification mechanism (SVM

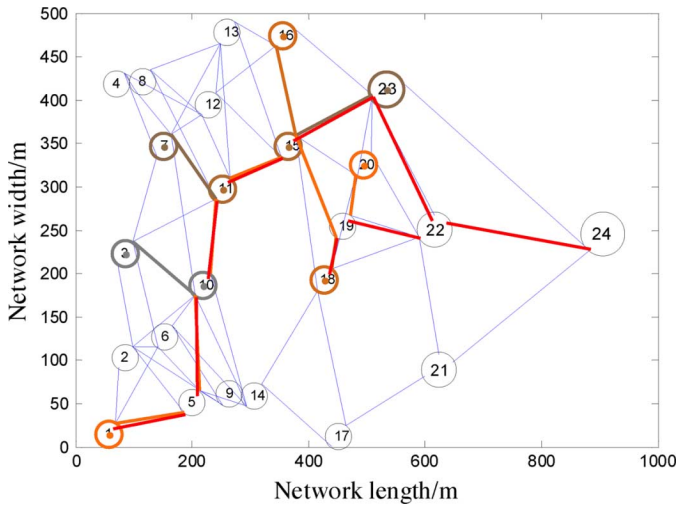


Fig. 9. Intrusion detection routing in home, neighborhood, and wide area network.

TABLE II  
THE INFLUENCING PARAMETERS AND THE OPTIMIZED VALUES

| CLONALG                    |        | AIRS2Parallel                                     |        |
|----------------------------|--------|---------------------------------------------------|--------|
| Parameters                 | Values | Parameters                                        | Values |
| CF (Clonal Factor)         | 0.9    | ATS (Affinity Threshold Scalar)                   | 0.4    |
| APS (Antibody Pool Size)   | 50     | CR (Clonal Rate)                                  | 10     |
| SPS (Selection Pool Size)  | 20     | HR (Hyper-mutation Rate)                          | 2      |
| TR (Total Replacements)    | 2      | kNN (Number of Nearest Neighbors)                 | 3      |
| NG (Number of Generations) | 10     | IMPS (Initial Memory Cell Pool Size)              | 50     |
| RPR (Remainder Pool Ratio) | 0.1    | NIAT (Number of Instances for Affinity Threshold) | -1     |
|                            |        | ST (Stimulation Threshold)                        | 0.5    |
|                            |        | TR (Total Resources)                              | 150    |

or AIS) to detect whether this set of features represents normal communications or an attack.

The classifiers are trained by using a specific portion of the NSL-KDD dataset in LIBSVM and the Weka plug-in for AIS algorithms. As these algorithms are nonlinear algorithms, the accuracy of the testing does not depend on the size of the training dataset. Rather, it depends on the proportions of the normal and the malicious records.

Our SVM model uses a Gaussian radial basis function for the kernel [28], a value of  $2^{15}$  for  $C$ , a value of  $2^3$  for  $\xi$ , and a value of 0.1 for  $\varepsilon$  as the parameters. The choice of the cache size (a parameter particular to LIBSVM) is 1600 MB because of the large scale of the training dataset.

Regarding the two AIS, the classification performance of the AIRS model is influenced by eight parameters and the performance of the CLONALG is influenced by six parameters [28]. The optimized values of the parameters which lead the algorithm to the best classification results are listed in Table II. The

choice of the cache size for operation of Weka is 2500 MB in order to prevent the out of memory errors.

We distribute 20 HAN IDS modules, 3 NAN IDS modules, and 1 WAN IDS module into the network nodes using both SVM and AIS algorithms. From the possible attacks analyzed in Section IV-A, the HAN IDS modules includes models for normal communications, DoS attacks, and probing attacks since these two types of attacks are more likely to occur. The NAN IDS modules include the best testing model of these HAN IDSs and the model trained for classifying the U2R attacks. For the module in the node of WAN IDS, besides the best HAN IDS model and the best NAN IDS model, it has the third model, which is used for classifying the R2L attacks.

As the process of simulating packet transmission and gathering data (such as packet type, connection duration, etc.) is complex, we have simplified it by including features of KDD99 dataset as a payload. It is this payload data that is classified as either normal or malicious.

If a source node has a communication with a destination node and the IDS in the destination node cannot clearly detect what kind of communication those features belong to, the features are then packaged and sent to a higher layer IDS for classification. Because of the limited possibility of types of attack samples, the IDSs in the lower layer can only detect limited types of intrusions as was stated in Section IV-A.

In any case, if the HAN IDS is unable to classify network traffic, then the data will be packaged and transmitted to a NAN IDS. If the NAN IDS is unable to classify traffic, it will first attempt to contact the closest NAN IDS for classification. The HAN IDS does not have this mechanism because of its smaller communication capacity and the limited detection ability. If the NAN IDS fails at classifying the data, it will then be sent on to the WAN IDS as it has the highest classification accuracy and ability. In Fig. 9 the search pattern for traffic classification is detailed. For example, the red line between nodes 19 and 22 represents the detection IDS as it is switched from HAN IDS to NAN IDS. The detection procedure is illustrated in Fig. 10.

The intrusion detection routing table is illustrated in Table III. The first column shows the destination nodes in Section IV-C that use the local IDS to make the detection and the last column represents the final IDSs used for the classification. The nodes between the destination nodes and the final nodes failed to perform a successful classification. The rule of optimal routing is also followed here. The first row of Table III illustrates the co-operation that can occur between the IDSs at multiple levels. In this case, the HAN IDS in node 10 then attempts to classify the communication as benign or malicious but is unable to do so. Because of this, node 10 packages and passes the communication up the network hierarchy to node 23 through the optimal routing of (10, 11, 15, 23). The NAN IDS in node 23 then attempts to classify the communication using its own AM, but it also fails to do so. The communication is once again packaged and passed to node 22 which has another NAN IDS. As this IDS cannot successfully classify the data, it will pass it up the hierarchy to node 24 with a WAN. If a successful classification is made, the classification will be transmitted down the hierarchy to the source node. The second and third rows show that the classifications succeed locally.

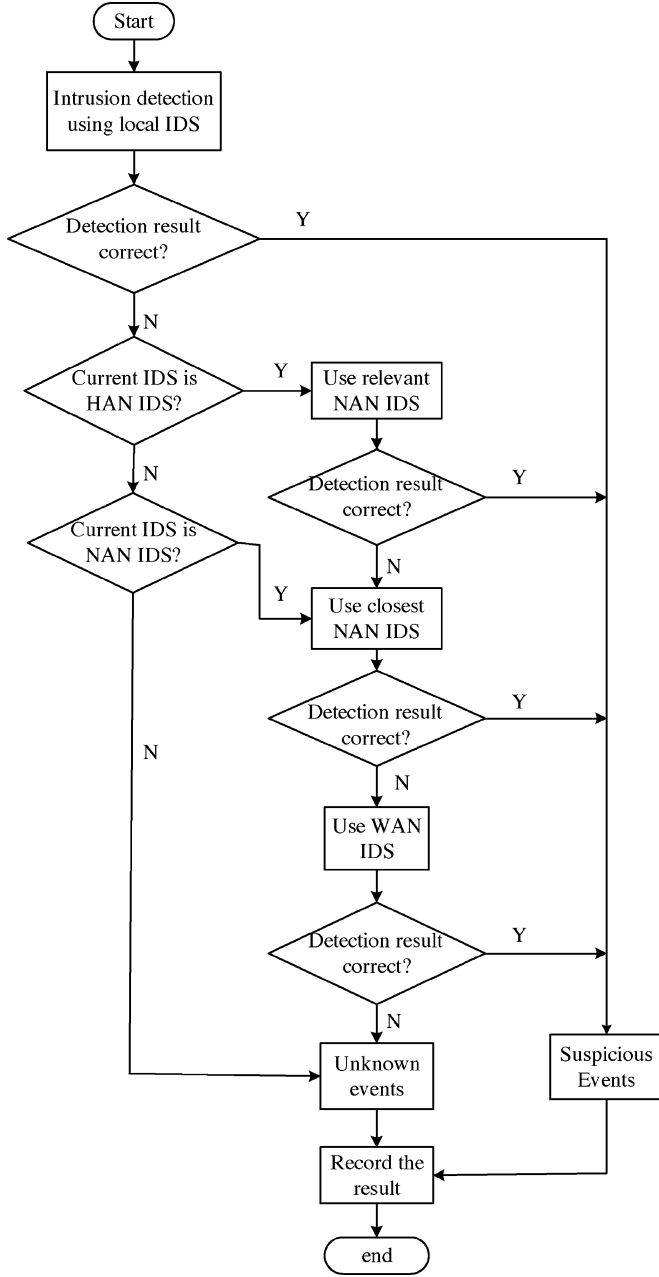


Fig. 10. Detection procedure in the SGDIDS.

TABLE III  
THE ROUTING TABLE OF THE COMMUNICATION AMONG NODES

| Destination node | Intermediate nodes |    |    |    | Final node |
|------------------|--------------------|----|----|----|------------|
| 10               | 11                 | 15 | 23 | 22 | 24         |
| 23               |                    |    |    |    | 23         |
| 15               |                    |    |    |    | 15         |
| 18               |                    |    | 19 |    | 22         |
| 1                | 5                  | 10 | 11 | 15 | 23         |

#### D. Results and Discussion

In the real smart grid communication network environment the transmission of packets will be continuous. This is simplified in our simulation as it is run a total of 50 times. Every time a record sample is randomly abstracted from the test dataset and sent to the IDS in the destination node, and the classification

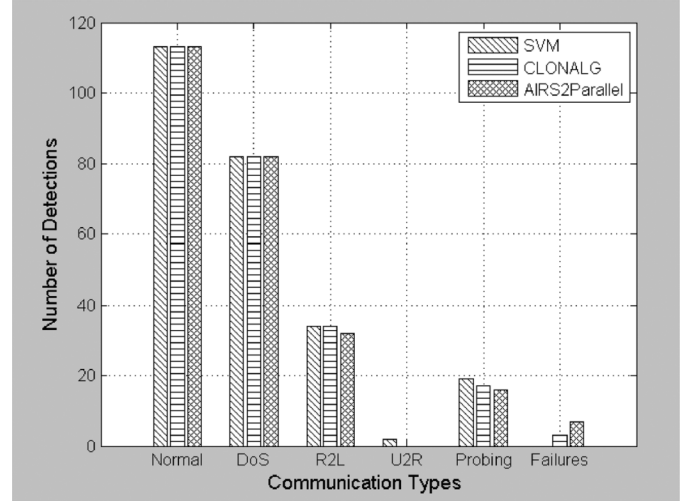


Fig. 11. Results using SVM and AIS algorithms.

occurs as described in Section IV-C, then a notice is displayed on the screen showing whether there is any threat to the network. The amount of time spent in classification, a count of the normal communications, a count of the total attacks, and the number of classification failures will be recorded and illustrated on the screen as well. If a classification cannot be properly made, a “detection failed” note will be posted and the total accuracy will be decreased.

In the simulation there are 113 normal communications, 82 DoS, 34 R2L, 2 U2R, and 19 probing attacks launched to the SGDIDS. The accumulation of the classification results and the classification failures using the SVM, CLONALG, and AIRS2Parallel algorithms from the simulation are recorded in Fig. 11. The first column shows the number of the normal communications followed by the accumulations of DoS, R2L, U2R, and probing attacks. The final column shows the number of times that classification fails. From Fig. 11, it can be found that the performance of the SVM is better than CLONALG and AIRS2Parallel algorithm at finding the R2L and U2R attacks. The classification failures of AIS algorithms are mostly due to insufficient attack samples in the training dataset. In particular, in roughly 126 000 records of training dataset only 995 are R2L and 52 are U2R. However, the SVM can classify them by using the appropriate proportion of data in the training dataset.

In order to achieve a better comparison of these algorithms, the overall false positive rate (FPR) and false negative rate (FNR) values of the SGDIDS are calculated by evaluating the whole test dataset. FPR values evaluate the cases where the normal communications are classified as attack scenarios and FNR evaluates the situations where the attacks of DoS, U2R, R2L, and probing are classified as normal communications and the type of attacks which they do not belong to. The records of the FPR and FNR are shown in Table IV.

From the values of the FPR/FNR in the Table IV it can be concluded that the SVM has stronger classification ability than the AIS algorithms. The main reason for the high FNR values of the AIS algorithms in the NAN and WAN layers is their failure to classify U2R attacks due to the low occurrence in the training dataset and their classification mechanism.

TABLE IV  
THE OVERALL FPR/FNR VALUES OF THE SGDIDS

| Classifier<br>Values | SVM   | CLONALG | AIRS2Parallel |
|----------------------|-------|---------|---------------|
| FPR                  | 0.67% | 0.7%    | 1.3%          |
| FNR                  | 2.15% | 21.02%  | 26.32%        |

## V. CONCLUSION AND FUTURE WORK

This paper proposes a new architecture, SGDIDS, in order to improve the cybersecurity of the smart grid by utilizing a hierarchical and distributed intrusion detection system in the wireless mesh network while also finding the optimal routing for smart grid communications. Security is improved via the classification of intrusion data using the SVM and AIS algorithms. The effectiveness of the new model for improving security is demonstrated through multiple simulations.

Though the two classifiers used in this study have exhibited acceptable or satisfactory performance, there is still room to further improve their performance. It is also possible that some other classifiers may be able to achieve better performance. Thus, one of our future research focuses is to evaluate the effectiveness of some existing classifiers for our problem. In this way, for each attack type the most effective classifier can be determined. As a result, the optimal combination of different classification techniques and different IDSs can be developed for achieving a higher overall accuracy for intrusion detection. In addition, an enhanced dataset derived based on the variations of power system parameters in the presence of cyberattacks will be developed for training and testing classifiers. The entire system will also be implemented in a SCADA testbed environment with real-time communications.

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Authors' photographs and biographies not available at the time of publication.